



Fig. 3: Basic functionalities in a commonly available True RMS multimeter (Credit: www.lensunsolar.com)

temperature coefficient represents the per cent change in these terms with a change in temperature ($\%/^{\circ}\text{C}$). This also shows how these values might deviate from their rated values in different climatic conditions.

Testing and measurement

Investing in proper photovoltaic testing is a good way of measuring how effectively the solar equipment is performing in the installed location before beginning with the operations.

Although testing individual components installed in solar energy plants most commonly deals with the measurement of current-voltage characteristics of solar panels and collectors, there are several other procedures too. These include measurement of the emission coefficient of materials, testing of PV modules in a humid environment, visual inspection and thermography of solar PV panels, testing of control units and control systems, performance testing, field testing, and many more.

The tools for measurements can range from simple digital multimeters that are easily available in labs to high-precision technique-based instruments. For test equipment that can characterise PV modules or source circuits, commissioning and safety

testers and portable I-V curve tracers are the most common. PVA-1500 PV Analyser Kit is a popular 1500-volt I-V curve tracer with built-in PV performance modelling and wireless irradiance, temperature, and tilt sensing.

Most of the professionals also carry an irradiance meter for checking real-time system performance and an infrared (IR) camera or thermometer with them these days. IR thermometers can take basic temperature measurements to do the same task as cameras, that is, identifying high-resistance electrical connections and thermally stressed overcurrent protection devices.

Among the handheld instruments, multimeters are a popular choice. Multimeter allows a user to test the voltage and current of any device, and therefore it can be used for testing solar panels. A switched multimeter is manually switched between the ranges to get an accurate reading of the selected function like the current, voltage, or other variables. Whereas an auto-range multimeter switches between ranges automatically. But both perform the same function.

To read the approved voltage of panels, check inside the converter box located at the back. Next, set the multimeter to have a higher reading than

the approved voltage. The reading of the multimeter must be as close as possible to the rating for high performance.

While performing the testing procedures, precautions must be taken so as not to cause any damage to the solar equipment. For example, the alligator clips attached between the panels and multimeter can only be disconnected after switching the multimeter off.

When connecting the controller and battery with the panel, first disconnect the panel from the controller, and then disconnect the battery from the controller. Similarly, during reconnection, first connect the controller to the battery and then the controller to the panel. Also, positive and negative connections must be duly taken care of.

Indian scenario

Solar energy companies in India are leaving no stone unturned to ensure the quality of their products in this highly competitive market. For instance, recently, Underwriters Laboratories (UL), a global safety certification organisation, announced that it has completed the PV module testing and certification for bifacial modules manufactured by Waaree Energies, a solar PV module manufacturer in India, as per the latest IEC standards.

According to a press release, Mukul Harne, AGM of Quality, Waaree Energies, said, "We partnered with UL due their long-standing understanding and experience in certification to IEC CB 2016 standards coupled with the harmonised UL 61730 standards that provide the US and international market access in one standard and the widespread acceptance of UL Mark in the US."

Testing labs have come up in the country in the past few years. ACME Solar Test Laboratory, a subsidiary company of ACME Cleantech Solutions, and a National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited lab, was established in Manesar, Haryana in 2017, which is engaged in the testing of crystalline silicon PV modules.

As the awareness about the importance of factors impacting cost systems and possibilities increases, the testing equipment and monitoring system market will grow further. **EFY**